Problem 1: Let A be a square matrix. Without using matrix entries, prove that $A + A^T$ is a symmetric matrix.

Solution

To prove that $A + A^T$ is symmetric, we need to show that $(A + A^T)^T = A + A^T$.

$$(A + A^T)^T = A^T + (A^T)^T.$$

Since the transpose of a transpose of a matrix is the matrix itself, we have $(A^T)^T = A$. Thus:

$$(A + A^T)^T = A^T + A.$$

Rearranging the terms:

$$(A + A^T)^T = A + A^T.$$

This shows that $A + A^T$ is equal to its transpose, and hence $A + A^T$ is symmetric.

Problem 2: By using elementary row operations, show that A^{-1} does not exist.

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

Solution

We apply the following row operations to simplify the matrix:
1. Perform $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 + R_1$:

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & -8 & -9 \\ 0 & 8 & 9 \end{bmatrix}.$$

2. Next, apply $R_3 \to R_3 + R_2$:

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & -8 & -9 \\ 0 & 0 & 0 \end{bmatrix}.$$

At this stage, the third row becomes all zeros, indicating that the matrix A is singular.

Problem 3: Let A be a square matrix. Without using matrix entries prove that $A - A^T$ is a skew-symmetric matrix.

Solution

To prove that $A - A^T$ is a skew-symmetric matrix, we need to show that:

$$(A - A^T)^T = -(A - A^T).$$

The transpose of $A - A^T$ is:

$$(A - A^T)^T = A^T - (A^T)^T.$$

Since the transpose of a transpose of a matrix is the matrix itself ($(A^T)^T = A$), we have:

$$(A - A^T)^T = A^T - A.$$

Observe that:

$$A^T - A = -(A - A^T).$$

Thus:

$$(A - A^T)^T = -(A - A^T).$$

Problem 4: Show that if A is an invertible matrix and $AB = AC$ then $B = C$.

Solution

The given equation is:

$$AB = AC.$$

Since A is invertible, A^{-1} exists. Multiply both sides of the equation $AB = AC$ by A^{-1} on the left:

$$A^{-1}(AB) = A^{-1}(AC).$$

Using the associativity of matrix multiplication, we can rewrite:

$$(A^{-1}A)B = (A^{-1}A)C.$$

Since $A^{-1}A = I$, where I is the identity matrix, the equation simplifies to:

$$IB = IC.$$

Multiplying any matrix by the identity matrix leaves it unchanged:

$$B = C.$$

Problem 5: By using elementary row operations, find A^{-1} if exist.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Solution

To find A^{-1}, we augment A with the identity matrix I and perform row operations to transform A into I, while applying the same operations on I.

Step 1: Augmented Matrix

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

Step 2: Row Operations

$R_2 \to R_2 + R_1$ and $R_3 \to R_3 - R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \end{array}\right].$$

$R_3 \to R_3 + R_2$ and $R_1 \to R_1 - R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right].$$

$R_2 \to R_2 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right].$$

Thus, the inverse of A is:

$$A^{-1} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Problem 6: Find elementary matrix E such that $EA = B$ for

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix}$$

Solution

To find the elementary matrix E, we observe that the third row of A is modified to produce B. By comparing the matrices A and B, we find that:

$$R_3 \to R_3 + 3R_1.$$

We construct E by applying the same row operation to the 3×3 identity matrix I:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Perform $R_3 \to R_3 + 3R_1$:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}.$$

Verification: Compute EA:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}.$$

$$EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix}.$$

Thus, $EA = B$.

Problem 7: Let $AX = B$ be a matrix equation of a system of linear equations. Prove that the system has unique solution if A is invertible matrix.

Solution

Given matrix equation of a linear system is:

$$AX = B,$$

Since A is invertible, A^{-1} exists. Left-multiply both sides of the equation by A^{-1}:

$$A^{-1}(AX) = A^{-1}B.$$

By the property of matrix multiplication, $A^{-1}A = I$. Thus, the equation becomes:

$$IX = A^{-1}B.$$

The identity matrix satisfies $IX = X$. Therefore:

$$X = A^{-1}B.$$

Uniqueness of the Solution
Since A^{-1} is unique and B is given, the product $A^{-1}B$ is uniquely determined. Thus, the system $AX = B$ has a unique solution:

$$X = A^{-1}B.$$

Problem 8: Let A be a 3×3 skew-symmetric matrix. Prove that the diagonal entries of A are all zero.

Solution

The property $A^T = -A$ implies:

$$a_{ij} = -a_{ji}, \quad \text{for all } 1 \leq i \leq 3, 1 \leq j \leq 3.$$

On the diagonal of A, the indices i and j are the same ($i = j$). Thus, for diagonal entries:

$$a_{ii} = -a_{ii}.$$

Rearranging the above equation:

$$2a_{ii} = 0.$$

This implies:

$$a_{ii} = 0, \quad \text{for } i = 1, 2, 3.$$

Problem 9: Find elementary matrix E such that $EA = B$ for

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$$

Solution

To find the elementary matrix E, we observe that A and B differ only in the third row:

$$R_3 \to R_3 - 2R_1.$$

Apply the row operation $R_3 \to R_3 - 2R_1$ to the identity matrix:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Perform $R_3 \to R_3 - 2R_1$:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

Step 2: Verify $EA = B$
Compute EA:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}.$$

Performing matrix multiplication gives:

$$EA = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}.$$

Thus, $EA = B$.

Problem 10: Find inverse of A if exist by using elementary row operations.

$$A = \begin{bmatrix} 2 & 0 \\ 6 & 5 \end{bmatrix}$$

Solution

To find A^{-1}, we augment A with the identity matrix I and perform row operations to transform A into I. The augmented matrix starts as:

$$\left[\begin{array}{cc|cc} 2 & 0 & 1 & 0 \\ 6 & 5 & 0 & 1 \end{array}\right].$$

Apply $R_1 \to \frac{1}{2}R_1$.

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & 0 \\ 6 & 5 & 0 & 1 \end{array}\right].$$

Apply $R_2 \to R_2 - 6R_1$:
The augmented matrix becomes:

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 5 & -3 & 1 \end{array}\right].$$

Apply $R_2 \to \frac{1}{5}R_2$.

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{-3}{5} & \frac{1}{5} \end{array}\right].$$

Hence

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{-3}{5} & \frac{1}{5} \end{bmatrix}.$$

Problem 11: Let A and B be two invertible matrices of same size. Prove that $(AB)^{-1} = B^{-1}A^{-1}$.

Solution

Using the associativity of matrix multiplication:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Also,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Hence,

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Problem 12: Find inverse of A if exist by using elementary row operations.

$$A = \begin{bmatrix} 1 & 0 \\ 3 & 5 \end{bmatrix}$$

Solution

To find A^{-1}, we augment A with the identity matrix I and perform row operations to transform A into I. The augmented matrix starts as:

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right].$$

Use $R_2 \to R_2 - 3R_1$:

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 5 & -3 & 1 \end{array}\right].$$

Apply $R_2 \to \frac{1}{5}R_2$.

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & -\frac{3}{5} & \frac{1}{5} \end{array}\right].$$

Hence,

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ -\frac{3}{5} & \frac{1}{5} \end{bmatrix}.$$

Problem 13: Prove that inverse of a non-singular matrix is unique.

Solution

Let B and C are two inverse of a non-singular matrix A. This implies:

$$BA = AB = I \text{ and } CA = AC = I$$

Using associative property of matrices we have:

$$C(AB) = (CA)B$$

$$CI = IB$$

Hence,

$$C = B$$

Problem 14: Given that matrix B is the inverse of A. Calculate $(A^3)^{-1}$.

$$A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$$

Solution

Since $(A^3)^{-1} = (A^{-1})^3$ and $A^{-1} = B$, so we need to calculate $(A^3)^{-1} = B^3$

$$B^2 = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = \begin{bmatrix} 25+14 & -10-6 \\ -35-21 & 14+9 \end{bmatrix} = \begin{bmatrix} 39 & -16 \\ -56 & 23 \end{bmatrix}$$

$$B^3 = B^2 \cdot B = \begin{bmatrix} 39 & -16 \\ -56 & 23 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = \begin{bmatrix} 195+112 & -78-48 \\ -280-161 & 112+69 \end{bmatrix} = \begin{bmatrix} 307 & -126 \\ -441 & 181 \end{bmatrix}$$

hence

$$(A^3)^{-1} = \begin{bmatrix} 307 & -126 \\ -441 & 181 \end{bmatrix}.$$

Problem 15: Given that matrix B is the inverse of A. Calculate $(A^3)^{-1}$.

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

Solution

Since $(A^3)^{-1} = (A^{-1})^3$ and $A^{-1} = B$, so we need to calculate $(A^3)^{-1} = B^3$

$$B^2 = B \cdot B = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}. = \begin{bmatrix} 9+2 & -6-2 \\ -3-1 & 2+1 \end{bmatrix} = \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix}$$

$$B^3 = B^2 \cdot B = \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 33+8 & -22-8 \\ -12-3 & 8+3 \end{bmatrix} = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix}$$

hence

$$(A^3)^{-1} = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix}.$$

Problem 16: Show that a 3×3 matrix with a column of zeros cannot have an inverse.

Solution

Let A be a 3×3 matrix with a column of zeros. Without loss of generality, assume the second column of A is all zeros:

$$A = \begin{bmatrix} a_{11} & 0 & a_{13} \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{bmatrix}.$$

Let $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$. Then the product BA is:

$$BA = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} a_{11} & 0 & a_{13} \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{bmatrix}.$$

Observe that:

$$\text{Second column of } BA = \begin{bmatrix} b_{11} \cdot 0 + b_{12} \cdot 0 + b_{13} \cdot 0 \\ b_{21} \cdot 0 + b_{22} \cdot 0 + b_{23} \cdot 0 \\ b_{31} \cdot 0 + b_{32} \cdot 0 + b_{33} \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

For $BA = I$, each column of BA must match the corresponding column of the identity matrix:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

However, as shown above, the second column of BA is: $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, which does not match the second column of I, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Therefore, $BA \neq I$ for any 3×3 matrix B.